IE 203 PS 6 – Solutions

Q1. Stationary Points of Single-Variable Functions

Solution

(a) $f(x) = xe^{-x}$

- **Stationary Points:** $f'(x) = e^{-x}(1-x) = 0 \implies x = 1$.
- **Local Classification:** $f''(x) = e^{-x}(x-2)$. Since $f''(1) = -e^{-1} < 0$, $x = 1$ is a **local maximum**.
- **Global Classification:**

$$\lim_{x\to-\infty} xe^{-x} = (-\infty)(e^{\infty}) = -\infty$$
$$\lim_{x\to\infty} xe^{-x} = \lim_{x\to\infty} \frac{x}{e^x} \overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{1}{e^x} = 0$$

  Since $\lim_{x\to-\infty} f(x) = -\infty$, there is **no global minimum**. Because the local maximum yields $f(1) = \frac{1}{e} > 0$ and the upper limit approaches 0, $x = 1$ is the **global maximum**.

(b) $f(x) = 6x^5 - 4x^3 + 10$

- **Stationary Points:** $f'(x) = 6x^2(5x^2 - 2) = 0 \implies x \in \left\{0, -\sqrt{\frac{2}{5}}, \sqrt{\frac{2}{5}}\right\}$.
- **Local Classification:** $f''(x) = 24x(5x^2 - 1)$.
  - $f''\left(\sqrt{\frac{2}{5}}\right) = 24\sqrt{\frac{2}{5}} > 0 \implies$ **local minimum**.
  - $f''\left(-\sqrt{\frac{2}{5}}\right) = -24\sqrt{\frac{2}{5}} < 0 \implies$ **local maximum**.
  - $f''(0) = 0$. Since $f'(x) < 0$ for $x \in \left(-\sqrt{\frac{2}{5}}, \sqrt{\frac{2}{5}}\right) \setminus \{0\}$, the sign of $f'(x)$ does not change around 0, so $x = 0$ is an **inflection point**.
- **Global Classification:**

$$\lim_{x\to-\infty}(6x^5 - 4x^3 + 10) = \lim_{x\to-\infty} 6x^5 = -\infty$$
$$\lim_{x\to\infty}(6x^5 - 4x^3 + 10) = \lim_{x\to\infty} 6x^5 = \infty$$

  Because the function's limits are unbounded in both directions, there is **neither a global maximum nor a global minimum**.

(c) $f(x) = (3x-2)^2(2x-3)^2$

- **Stationary Points:**

$$\begin{aligned} f'(x) &= 2(3x-2)(3)(2x-3)^2 + (3x-2)^2(2)(2x-3)(2) \\ &= 2(3x-2)(2x-3)[3(2x-3) + 2(3x-2)] \\ &= 2(3x-2)(2x-3)(12x-13) \end{aligned}$$

  Setting $f'(x) = 0 \implies x \in \left\{\frac{2}{3}, \frac{13}{12}, \frac{3}{2}\right\}$.

- **Local Classification:** By the First Derivative Test:

- $f'(x)$ changes from $-$ to $+$ at $\frac{2}{3} \implies$ **local minimum**.
- $f'(x)$ changes from $+$ to $-$ at $\frac{13}{12} \implies$ **local maximum**.
- $f'(x)$ changes from $-$ to $+$ at $\frac{3}{2} \implies$ **local minimum**.

- **Global Classification:** We formally evaluate the limits:

$$\lim_{x\to-\infty}(3x-2)^2(2x-3)^2 = \lim_{x\to-\infty} 36x^4 = \infty$$
$$\lim_{x\to\infty}(3x-2)^2(2x-3)^2 = \lim_{x\to\infty} 36x^4 = \infty$$

Because both limits approach ∞, there is **no global maximum**. Because the local minimums yield $f\left(\frac{2}{3}\right) = f\left(\frac{3}{2}\right) = 0$ and the extreme limits approach ∞, both $x = \frac{2}{3}$ and $x = \frac{3}{2}$ are **global minimums**.

Q2. Gradient, Hessian, and Convexity of a Quadratic

Solution

(a)

$$\nabla f(x_1, x_2) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 6x_1 + 2x_2 - 4 \\ 2x_1 + 4x_2 + 6 \end{bmatrix}$$

$$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$$

(b) Characteristic equation $\det(H - \lambda I) = 0$:

$$\begin{aligned} \det \begin{bmatrix} 6-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix} &= (6-\lambda)(4-\lambda) - 4 \\ &= \lambda^2 - 10\lambda + 20 = 0 \end{aligned}$$

$$\implies \lambda_{1,2} = \frac{10 \pm \sqrt{100-80}}{2} = 5 \pm \sqrt{5}$$

Since $\lambda_1 = 5 + \sqrt{5} > 0$ and $\lambda_2 = 5 - \sqrt{5} > 0$, the Hessian matrix is positive definite globally. Thus, $f(x_1, x_2)$ is **strictly convex** on $\mathbb{R}^2$.

Q3. Classifying Convexity of Functions

Solution

(a) Both: $f(x) = 2x + 5$ is affine. Because $f''(x) = 0$ for all $x \in \mathbb{R}$, it globally satisfies both the convex ($f''(x) \geq 0$) and concave ($f''(x) \leq 0$) inequality definitions.

(b) Neither: $f(x) = x^3 \implies f''(x) = 6x$. Since $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$, the function changes concavity across the origin, thereby acting as neither globally.

(c) Convex: $f(x) = x^3$ on restricted domain $[0, \infty)$. $f''(x) = 6x \geq 0$ exclusively over this non-negative interval.

(d) Concave: $f(x) = \ln(x)$ on $(0, \infty)$. $f''(x) = -\frac{1}{x^2} < 0$ globally for all $x \in (0, \infty)$.

(e) Convex: $f(x) = x\ln(x)$ on $(0, \infty)$. $f'(x) = \ln(x) + 1 \implies f''(x) = \frac{1}{x} > 0$ globally for all $x \in (0, \infty)$.

(f) Convex: $f(x) = \frac{1}{x}$ on $(0, \infty)$. $f'(x) = -x^{-2} \implies f''(x) = 2x^{-3} > 0$ globally for all $x \in (0, \infty)$.

(g) Convex: Proof using the formal algebraic definition. Let $\lambda \in [0, 1]$ and $x, y \in \mathbb{R}$.

$$\begin{aligned} f(\lambda x + (1-\lambda)y) &= g(\lambda x + (1-\lambda)y) + 3h(\lambda x + (1-\lambda)y) \\ &\leq [\lambda g(x) + (1-\lambda)g(y)] + 3[\lambda h(x) + (1-\lambda)h(y)] \quad \text{(since } g, h \text{ are convex)} \\ &= \lambda[g(x) + 3h(x)] + (1-\lambda)[g(y) + 3h(y)] \quad \text{(rearranging terms algebraically)} \\ &= \lambda f(x) + (1-\lambda)f(y) \end{aligned}$$

Therefore, the inequality holds globally, retaining convexity.

(h) Convex: Method 1 (Formal λ Definition): Let $\lambda \in [0, 1]$ and $x, y \in \mathbb{R}$.

$$\begin{aligned} f(\lambda x + (1-\lambda)y) &= \max\{g(\lambda x + (1-\lambda)y), h(\lambda x + (1-\lambda)y)\} \\ &\leq \max\{\lambda g(x) + (1-\lambda)g(y), \lambda h(x) + (1-\lambda)h(y)\} \quad \text{(by convexity of } g \text{ and } h) \\ &\leq \max\{\lambda f(x) + (1-\lambda)f(y), \lambda f(x) + (1-\lambda)f(y)\} \quad \text{(since } g \leq f \text{ and } h \leq f) \\ &= \lambda f(x) + (1-\lambda)f(y) \quad \text{(simplifying identical nested arguments)} \end{aligned}$$

Method 2 (Topological Epigraph Analysis): A function is convex if and only if its epigraph (the region vertically at or above the graph) is a convex set. By definition, $\text{epi}(f) = \{(x, t) \in \mathbb{R}^2 \mid t \geq f(x)\}$. Since $f(x) = \max\{g(x), h(x)\}$, a point (x, t) belongs strictly to $\text{epi}(f)$ if and only if $t \geq g(x)$ and $t \geq h(x)$. Consequently, $\text{epi}(f) = \text{epi}(g) \cap \text{epi}(h)$. Because g and h are independently convex functions, their respective epigraphs are convex sets. Since the intersection of any collection of convex sets is universally a convex set, $\text{epi}(f)$ is convex.

(i) Neither (in general):
Topological Analysis: The epigraph of the pointwise minimum is the inclusive union of individual epigraphs: $\text{epi}(f) = \text{epi}(g) \cup \text{epi}(h)$. Since the mathematical union of two convex sets does not guarantee a convex set, global convexity is logically inconclusive. Similarly, its hypograph is the mathematical intersection of respective hypographs: $\text{hypo}(f) = \text{hypo}(g) \cap \text{hypo}(h)$. While the intersection of convex sets is indeed convex, the hypograph of a strictly convex function is inherently *not* a convex set, leaving global concavity equally inconclusive.

Explicit Counterexamples ($\lambda = 0.5$): Let $g(x) = x^2$ and $h(x) = (x-2)^2$.

- *Not Convex:* Evaluate at $x_1 = 0$ and $x_2 = 2$. $f(0) = 0$ and $f(2) = 0$. At the mathematical midpoint, $f(1) = 1$. The foundational convexity condition $f(1) \leq 0.5f(0) + 0.5f(2) \implies 1 \leq 0$ explicitly fails.
- *Not Concave:* Evaluate closely at $x_1 = 0$ and $x_2 = 1$. $f(0) = 0$ and $f(1) = 1$. At the internal midpoint, $f(0.5) = 0.25$. The foundational concavity condition $f(0.5) \geq 0.5f(0) + 0.5f(1) \implies 0.25 \geq 0.5$ explicitly fails.

Q4. Three-Variable Hessian and Convexity

Solution

(a) The first-order gradient and second-order Hessian matrix are derived systematically:

$$\nabla f(x_1, x_2, x_3) = \begin{bmatrix} 4x_1 + 4x_2 - 3 \\ 4x_1 + 10x_2 + 2 \\ 6x_3 - 1 \end{bmatrix}$$

$$H(x_1, x_2, x_3) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 4 & 4 & 0 \\ 4 & 10 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

(b) Characteristic equation $\det(H - \lambda I) = 0$:

$$\begin{aligned}
\det \begin{bmatrix} 4-\lambda & 4 & 0 \\ 4 & 10-\lambda & 0 \\ 0 & 0 & 6-\lambda \end{bmatrix} &= 0 \\
(6-\lambda) \cdot \det \begin{bmatrix} 4-\lambda & 4 \\ 4 & 10-\lambda \end{bmatrix} &= 0 \quad \text{(Laplace expansion along the 3rd row)} \\
(6-\lambda)\left[(4-\lambda)(10-\lambda) - 4(4)\right] &= 0 \\
(6-\lambda)(\lambda^2 - 14\lambda + 24) &= 0 \\
(6-\lambda)(\lambda-12)(\lambda-2) &= 0
\end{aligned}$$

Thus, the three characteristic roots are $\lambda_1 = 6$, $\lambda_2 = 12$, and $\lambda_3 = 2$.

(c) Since all three eigenvalues are strictly positive ($\lambda_{1,2,3} > 0$), the Hessian matrix restricts to being positive definite globally. Therefore, the function $f(x_1, x_2, x_3)$ is **strictly convex** across $\mathbb{R}^3$.

Q5. Local Convexity and Convexity Region

Solution

(a)

$$\begin{aligned}
\nabla f(x,y) &= \begin{bmatrix} 4x^3 + 3y \\ 3x + 2y \end{bmatrix} \\
H(x,y) &= \begin{bmatrix} 12x^2 & 3 \\ 3 & 2 \end{bmatrix}
\end{aligned}$$

(b) At $(1,1)$, we evaluate $H(1,1) = \begin{bmatrix} 12 & 3 \\ 3 & 2 \end{bmatrix}$. Characteristic equation $\det(H - \lambda I) = 0$:

$$\begin{aligned}
(12-\lambda)(2-\lambda) - 9 &= 0 \\
\lambda^2 - 14\lambda + 15 &= 0 \\
\implies \lambda_{1,2} &= \frac{14 \pm \sqrt{196 - 60}}{2} = 7 \pm \sqrt{34}
\end{aligned}$$

Since $34 < 49$, we strictly know $\sqrt{34} < 7$, which guarantees the lower bound $\lambda_2 = 7 - \sqrt{34} > 0$. With both eigenvalues definitively positive, $H(1,1)$ is positive definite, making $f(x,y)$ **locally strictly convex** at $(1,1)$.

(c) For $f(x,y)$ to be convex over a spatial region, $H(x,y)$ must be positive semidefinite. We evaluate all principal minors:

- 1×1 minors: $M_1 = 12x^2 \geq 0$ and $M_2 = 2 \geq 0$ trivially globally.
- 2×2 minor: $\det(H) = 24x^2 - 9 \geq 0 \implies x^2 \geq \frac{3}{8}$.

Thus, the specific region of convexity is mathematically bound by:

$$\mathcal{S} = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; |x| \geq \sqrt{\frac{3}{8}} \right\}$$

Q6. Convexity Analysis of Multivariate Functions

Solution

Note: While evaluating either the principal minors or the eigenvalues is mathematically sufficient to determine convexity natively, we generally provided both approaches here for algebraic completeness. We

evaluated Leading Principal Minors (LPMs) using Sylvester's criterion first. For definitive structural tests (e.g., proving strictly positive/negative definite matrices or strict indefiniteness), checking just the Leading Principal Minors is exclusively sufficient, and we will ***not*** *proceed to calculate the remaining non-leading PMs explicitly. However, if the LPMs are fundamentally inconclusive (such as evaluating to zero, opening paths to semidefiniteness), we will explicitly evaluate all remaining principal minors.*

(a) $f(x_1, x_2) = x_1^3 + 2x_2^2$

$$\nabla f(x) = \begin{bmatrix} 3x_1^2 \\ 4x_2 \end{bmatrix}, \quad H(x) = \begin{bmatrix} 6x_1 & 0 \\ 0 & 4 \end{bmatrix}$$

- **LPMs:** $D_1 = 6x_1$ and $D_2 = 24x_1$. The signs depend structurally on the coordinate x_1. For instance, evaluating at any $x_1 < 0$ mathematically guarantees $D_2 < 0$. Because an even-order LPM evaluates strictly negative, the matrix is definitively indefinite. Since LPMs are uniquely sufficient in proving strict indefiniteness, we do not generate other PMs.
- **Eigenvalues:** Because H is a true diagonal matrix, its eigenvalues echo its diagonal entries natively: $\lambda_1 = 6x_1$ and $\lambda_2 = 4$. Because λ_1 can cross negative while λ_2 is strictly positive, the root bounds have mixed signs, formally echoing an indefinite matrix.
- **Conclusion:** The function is globally **neither convex nor concave**.

(b) $f(x_1, x_2) = -x_1^2 - 5x_2^2 + 2x_1x_2 + 10x_1 - 10x_2$

$$\nabla f(x) = \begin{bmatrix} -2x_1 + 2x_2 + 10 \\ 2x_1 - 10x_2 - 10 \end{bmatrix}, \quad H(x) = \begin{bmatrix} -2 & 2 \\ 2 & -10 \end{bmatrix}$$

- **LPMs:** $D_1 = -2 < 0$ and $D_2 = (-2)(-10) - (2)^2 = 16 > 0$. Because the continuous minors strictly cycle $(-, +)$ leading with a strictly bound zero, H is definitively negative definite. Since LPMs algebraically guarantee structural definiteness, evaluating other PMs is unnecessary.
- **Eigenvalues:** Solving $\det(H - \lambda I) = \lambda^2 + 12\lambda + 16 = 0$ yields algebraic roots corresponding to $\lambda_{1,2} = -6 \pm 2\sqrt{5}$. Since $2\sqrt{5} = \sqrt{20} < \sqrt{36} = 6$, we inherently know $2\sqrt{5} < 6$, isolating both radical roots rigidly into $\lambda_{1,2} < 0$. The roots unequivocally confirm negative definiteness.
- **Conclusion:** The function is **strictly concave** globally on $\mathbb{R}^2$.

(c) $f(x_1, x_2, x_3) = x_1x_2 + 2x_1^2 + x_2^2 + 2x_3^2 - 6x_1x_3$

$$\nabla f(x) = \begin{bmatrix} 4x_1 + x_2 - 6x_3 \\ x_1 + 2x_2 \\ -6x_1 + 4x_3 \end{bmatrix}, \quad H(x) = \begin{bmatrix} 4 & 1 & -6 \\ 1 & 2 & 0 \\ -6 & 0 & 4 \end{bmatrix}$$

- **LPMs:** We explicitly sequence the leading principal determinants:

$$\begin{aligned} D_1 &= 4 > 0 \\ D_2 &= \det \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} = 8 - 1 = 7 > 0 \\ D_3 &= 4(8 - 0) - 1(4 - 0) - 6(0 + 12) = 32 - 4 - 72 = -44 < 0 \end{aligned}$$

  Because $D_1 > 0$, the structure definitively fails negative definiteness. However, because $D_3 < 0$, it strictly shatters the positivity pattern required for positive definiteness. Thus, H is rigidly indefinite. Since LPMs logically dictate indefiniteness cleanly, calculating alternative individual PM branches is mathematically unnecessary.

(d) $f(x_1, x_2) = x_1 e^{-(x_1+x_2)}$

$$\nabla f(x) = e^{-(x_1+x_2)} \begin{bmatrix} 1 - x_1 \\ -x_1 \end{bmatrix}, \quad H(x) = e^{-(x_1+x_2)} \begin{bmatrix} x_1 - 2 & x_1 - 1 \\ x_1 - 1 & x_1 \end{bmatrix}$$

- **LPMs:** Defining the generic coefficient $E = e^{-(x_1+x_2)} > 0$, we find $D_1 = E(x_1 - 2)$ alongside $D_2 = E^2[x_1(x_1-2)-(x_1-1)^2] = E^2(x_1^2-2x_1-x_1^2+2x_1-1) = -E^2$. Regardless of point limits, since

squared exponential factors confirm $E^2 > 0$ strictly on $\mathbb{R}^2$, the determinant $D_2 = -E^2 < 0$ always holds. Tracing an even-order leading principal minor as structurally evaluating strictly negative natively overrides any initial boundary logic, stripping the matrix instantly to indefinite regardless of D_1. Because LPM boundaries are flawlessly sufficient for indefinite analysis, supplementary PM mapping skips.

- **Eigenvalues:** Formulating $\det(H - \lambda I) = 0 \implies \lambda^2 - E(2x_1 - 2)\lambda - E^2 = 0$. By the quadratic sequence, $\lambda_{1,2} = E[(x_1 - 1) \pm \sqrt{(x_1 - 1)^2 + 1}]$. Because adding internal increments strictly guarantees $\sqrt{(x_1 - 1)^2 + 1} > |x_1 - 1|$, the separated branch limits must bifurcate exclusively into one universally positive root isolated against one universally negative root. The conflicting signs reflect indefiniteness.

- **Conclusion:** The function is globally **neither convex nor concave**.

(e) $f(x_1, x_2) = (x_1 + x_2)^2$

$$\nabla f(x) = \begin{bmatrix} 2(x_1 + x_2) \\ 2(x_1 + x_2) \end{bmatrix}, \quad H(x) = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

- **LPMs:** Evaluating $D_1 = 2 > 0$ mapped alongside $D_2 = \det(H) = 4 - 4 = 0$. Because bounding $D_2 = 0$ violates Sylvester's explicit boundaries requiring rigorous distinct > 0 metrics for positive definiteness, evaluating only leading determinants stalls identically **inconclusive**. Logically mapping the underlying semidefiniteness necessitates investigating the secondary PMs directly.

- **PMs (All Remaining Principal Minors):**

$$\begin{aligned} \text{1st-order PMs:} \quad & M_1 = 2 \geq 0, \quad M_2 = 2 \geq 0 \\ \text{2nd-order PM:} \quad & M_{12} = \det(H) = 0 \geq 0 \end{aligned}$$

  Because every exhaustive combination factor of principal minors computes reliably to generic non-negative figures, substituting guarantees the Hessian acts positive semidefinite.

- **Eigenvalues:** Analyzing the root matrix polynomial $(2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = 0 \implies \lambda_{1,2} \in \{0, 4\}$. Bounding generic eigenvalue calculations rigidly against zero sets $\lambda_i \geq 0$, identifying natively positive semidefiniteness correctly.

- **Conclusion:** The function is **convex** globally on $\mathbb{R}^2$ (but not strictly convex).

Q7. Parametric Convexity

Solution

$$\nabla f(x, y) = \begin{bmatrix} 4x^3 + \alpha y \\ \alpha x + 4y^3 \end{bmatrix}, \quad H(x, y) = \begin{bmatrix} 12x^2 & \alpha \\ \alpha & 12y^2 \end{bmatrix}$$

(a) Evaluating the parametric Hessian algebraically at the coordinate $(1, 1)$ yields:

$$H(1, 1) = \begin{bmatrix} 12 & \alpha \\ \alpha & 12 \end{bmatrix}$$

For strict local convexity, $H(1, 1)$ must evaluate accurately as strictly positive definite. We calculate the leading principal minors:

- $D_1 = 12 > 0$ (verified independently of α)
- $D_2 = \det(H) = 144 - \alpha^2 > 0$

Solving the secondary determinant constraint structurally imposes $\alpha^2 < 144$, which equates directly to the scalar interval bounded by $\alpha \in (-12, 12)$. Thus, the function is strictly convex at $(1, 1)$ selectively for $\alpha \in (-12, 12)$.

(b) For valid global convexity spanning $\mathbb{R}^2$, the underlying Hessian matrix $H(x, y)$ must remain consistently positive semidefinite. Inherently, the determinant constraint $\det(H) = 144x^2y^2 - \alpha^2 \geq 0$ must hold seamlessly across all possible coordinate networks without failure.

Applying this rigorous boundary check specifically at the origin $(0, 0)$ forces:

$$\begin{aligned} 144(0)^2(0)^2 - \alpha^2 &\geq 0 \\ -\alpha^2 &\geq 0 \end{aligned}$$

The only real algebraic scalar dynamically capable of fulfilling $-\alpha^2 \geq 0$ is exactly $\alpha = 0$. Substituting $\alpha = 0$ directly into the generic Hessian physically isolates an independent diagonal matrix $H(x, y) = \begin{bmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{bmatrix}$. Because its core principal minors continuously track to $M_{11} = 12x^2 \geq 0$, $M_{22} = 12y^2 \geq 0$, and $\det(H) = 144x^2y^2 \geq 0$ across the global domain, substituting $\alpha = 0$ uniquely and successfully locks the function into global convexity.

Q8. FONC and Indefinite Hessian

Solution

(a) The First-Order Necessary Condition (FONC) $\nabla f(x, y) = \mathbf{0}$ generates the coupled algebraic system directly:

$$\nabla f(x, y) = \begin{bmatrix} 2x - e^{x-y} + e^{y-x} \\ e^{x-y} - e^{y-x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Isolating the secondary equation securely yields $e^{x-y} = e^{y-x} \implies x - y = y - x$, enforcing symmetric coordinates $x = y$. Substituting this structural constraint back into the primary equation equates $2x - e^0 + e^0 = 0 \implies 2x = 0$, isolating precisely $x = y = 0$. Thus, the exact origin $(0, 0)$ uniquely satisfies the critical stationary condition globally.

(b) The second-order cross-partial derivatives establish the formal mapped Hessian matrix $H(x, y)$:

$$H(x, y) = \begin{bmatrix} 2 - (e^{x-y} + e^{y-x}) & e^{x-y} + e^{y-x} \\ e^{x-y} + e^{y-x} & -(e^{x-y} + e^{y-x}) \end{bmatrix}$$

Defining the generic substitution scalar $A = e^{x-y} + e^{y-x}$, we observe that because exponential functions are unconditionally strictly positive, $A > 0$ natively. (Furthermore, structurally either the exponent $(x - y) \geq 0$ or $(y - x) \geq 0$, explicitly guaranteeing at least one $e^{\geq 0}$ component inherently must evaluate to ≥ 1, cementing natively that $A \geq 1$).

$$H(x, y) = \begin{bmatrix} 2 - A & A \\ A & -A \end{bmatrix}$$

Isolating the characteristic bounding equation structurally computes $\det(H - \lambda I) = 0$:

$$\begin{aligned} \det \begin{bmatrix} 2 - A - \lambda & A \\ A & -A - \lambda \end{bmatrix} &= 0 \\ (2 - A - \lambda)(-A - \lambda) - A^2 &= 0 \\ -2A - 2\lambda + A^2 + 2A\lambda + \lambda^2 - A^2 &= 0 \\ \lambda^2 + (2A - 2)\lambda - 2A &= 0 \end{aligned}$$

Applying the explicit quadratic formula successfully traces their characteristic boundary roots:

$$\begin{aligned}
\lambda_{1,2} &= \frac{-(2A-2) \pm \sqrt{(2A-2)^2 - 4(1)(-2A)}}{2} \\
&= \frac{-2A+2 \pm \sqrt{4A^2 - 8A + 4 + 8A}}{2} \\
&= \frac{-2A+2 \pm 2\sqrt{A^2+1}}{2} \\
&= 1 - A \pm \sqrt{A^2+1}
\end{aligned}$$

We sequentially map the explicit limit orientations of these structural roots independent of (x, y):

- **First Root (λ_1):** Because mathematically $\sqrt{A^2+1} > \sqrt{A^2} = A$, the inequality isolating the positive bound $\sqrt{A^2+1} - A > 0$ strictly anchors. The root $\lambda_1 = 1 + (\sqrt{A^2+1} - A) > 1$ globally guarantees the eigenvalue is rigorously **positive**.
- **Second Root (λ_2):** Since established logically that $A \geq 1$, the leading scalar structure analytically equates $1 - A \leq 0$. Since subtracting the positive geometric radical strictly diminishes this constraint further into extreme negative territory, the eigenvalue λ_2 traces inherently **negative**.

Because the explicitly mapped Hessian eigenvalues evaluate cleanly to absolute mixed signs unconditionally ($\lambda_1 > 0$ alongside isolated $\lambda_2 < 0$), $H(x, y)$ acts rigidly **indefinite**. Consistently evaluating an indefinite Hessian structure dictates the parent function $f(x, y)$ sits uniquely **neither convex nor concave** generally.

Q9. Formal Convexity Proofs

Solution

(a) Let $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$. We verify the convexity inequality directly using the core geometric properties of vector norms:

$$\begin{aligned}
f(\lambda x + (1-\lambda)y) &= \|\lambda x + (1-\lambda)y\| \\
&\leq \|\lambda x\| + \|(1-\lambda)y\| \quad \text{(Triangle Inequality)} \\
&= |\lambda|\|x\| + |1-\lambda|\|y\| \quad \text{(Absolute Homogeneity)} \\
&= \lambda\|x\| + (1-\lambda)\|y\| \quad \text{(since } \lambda, 1-\lambda \geq 0) \\
&= \lambda f(x) + (1-\lambda) f(y)
\end{aligned}$$

Thus, any valid vector norm rigorously satisfies the fundamental definition of a convex function.

(b) Let $(x_1, t_1), (x_2, t_2) \in \mathrm{dom}(f)$ where $t_1, t_2 > 0$, and let $\lambda \in [0, 1]$. We formally map the convex combination of the parametric scalar denominators as $t_\lambda = \lambda t_1 + (1-\lambda)t_2 > 0$. Evaluating the perspective function yields:

$$f(\lambda x_1 + (1-\lambda)x_2,\, t_\lambda) = t_\lambda \cdot h\left(\frac{\lambda x_1 + (1-\lambda)x_2}{t_\lambda}\right)$$

We dynamically deduce a structural inner weight $\alpha = \frac{\lambda t_1}{t_\lambda}$. Because $t_1, t_2 > 0$ and $\lambda \in [0, 1]$, it strictly anchors $\alpha \in [0, 1]$. By symmetric deduction, $1 - \alpha = \frac{(1-\lambda)t_2}{t_\lambda}$. Factoring the internal argument into this

mapped scalar combination generates a continuous algebraic bounds progression:

$$
\begin{aligned}
f(\lambda x_1 + (1-\lambda)x_2,\, t_\lambda) &= t_\lambda \cdot h\left(\alpha \frac{x_1}{t_1} + (1-\alpha)\frac{x_2}{t_2}\right) \\
&\leq t_\lambda \left[\alpha\, h\left(\frac{x_1}{t_1}\right) + (1-\alpha)\, h\left(\frac{x_2}{t_2}\right)\right] \quad \text{(by convexity of } h\text{)} \\
&= t_\lambda \left[\frac{\lambda t_1}{t_\lambda}\, h\left(\frac{x_1}{t_1}\right) + \frac{(1-\lambda)t_2}{t_\lambda}\, h\left(\frac{x_2}{t_2}\right)\right] \quad \text{(substituting } \alpha\text{)} \\
&= \lambda t_1\, h\left(\frac{x_1}{t_1}\right) + (1-\lambda)t_2\, h\left(\frac{x_2}{t_2}\right) \\
&= \lambda f(x_1, t_1) + (1-\lambda) f(x_2, t_2)
\end{aligned}
$$

This elegantly preserves rigorous global convexity for all generic perspective transformations.

(c) *Discarded.*